PROJECT CONTROLS FORMULAE IN STANDARD PRACTICE

1. EARNED VALUE MANAGEMENT (EVM)

PV (Planned Value)	Budgeted cost of work scheduled
EV (Earned Value)	Budgeted cost of work performed
AC (Actual Cost)	Actual cost of work performed
CV (Cost Variance)	$CV = EV - AC$
SV (Schedule Variance)	$SV = EV - PV$
CPI (Cost Performance Index)	$CPI = \frac{EV}{AC}$
SPI (Schedule Performance Index)	$SPI = \frac{EV}{PV}$

EAC (Estimate at Completion)

a) $EAC = BAC / CPI$

b) $EAC = AC + (BAC - EV)/CPI$

ETC (Estimate to Complete)

a) $ETC = EAC - AC$

b) $ETC = (BAC - EV)/CPI$

VAC (Variance at Completion) $VAC = BAC - EAC$

TCPI (To Complete Performance Index)

a) $TCPI = (BAC - EV)/(BAC - AC)$

b) $TCPI = (BAC - EV)/ETC$

BAC (Budget at Completion) BAC = Total budgeted cost

2. SCHEDULE MANAGEMENT

Forward Pass

ES (Early Start) $ES_i = \max(EF \text{ of all predecessors})$

EF (Early Finish) $EF_i = ES_i + Duration_i$

Backward Pass

LF (Late Finish) $LF_i = \min(LS \text{ of all successors})$

LS (Late Start) $LS_i = LF_i - Duration_i$

Float Calculations

Total Float (TF) $TF_i = LS_i - ES_i = LF_i - EF_i$

Free Float (FF) $FF_i = \min(ES \text{ of all successors}) - EF_i$

Independent Float (IF) $IF_i = \max(0, \min(ES \text{ of all successors}) - \max(EF \text{ of all predecessors}) - Duration_i)$

Critical Path

Path with zero total float

Project Duration = max(EF) of all activities

Note: i = activity

3. COST & RESOURCE MANAGEMENT

Cost Variance

$$CV = EV - AC$$

Cost Performance Index

$$CPI = \frac{EV}{AC}$$

Budget Variance

$$BV = BAC - EV$$

Estimate at Completion

$$EAC = AC + \frac{(BAC - EV)}{CPI} \quad \text{or} \quad EAC = \frac{BAC}{CPI}$$

Estimate to Complete

$$ETC = EAC - AC$$

To Complete Performance Index

$$TCPI = \left(\frac{(BAC - EV)}{(BAC - AC)}\right) \quad \text{or} \quad TCPI = \left(\frac{(BAC - EV)}{ETC}\right)$$

Resource Utilization

$$\text{Resource Utilization (\%)} = \left(\frac{\text{Actual Units}}{\text{(Available Units)}}\right) \times 100$$

4. QUALITY MANAGEMENT

Defect Density (DD)

$$DD = \frac{\text{Number of Defects}}{\text{Size of Product}}$$

Defects Per Million Opportunities (DPMO)

$$DPMO = \left(\frac{\text{(Number of Defects)}}{\text{Opportunities}}\right) \times 1{,}000{,}000$$

Yield Percentage

$$\text{Yield (\%)} = \left(\frac{\text{(Opportunities - Defects)}}{\text{Opportunities}}\right) \times 100$$

Cost of Quality (CoQ)

CoQ = Prevention Cost + Appraisal Cost + Failure Cost

Failure Cost = Internal Failure Cost + External Failure Cost

5. RISK MANAGEMENT

Risk Exposure (RE)

RE = Probability of Occurrence × Impact

Expected Monetary Value (EMV)

$$EMV = \sum(\text{Probability}_i \times \text{Impact}_i)$$

Decision Tree – Expected Value (EV)

$$EV = \sum(\text{Probability of Path} \times \text{Payoff of Path})$$

Monte Carlo Simulation

Use probability distribution to compute mean, standard deviation, $P(X \le x)$, $P(X \ge x)$

Risk Priority Number (RPN)

RPN = Probability × Impact × Detectability

6. PROCUREMENT MANAGEMENT

Total Cost of Ownership (TCO)

TCO = Purchase Price + Operating Cost + Maintenance Cost + Disposal Cost

Make or Buy Analysis

Net Benefit = (Cost of Purchase – Cost to Make)

Choose option with higher net benefit

Seller Performance Index (SPI_p)

$$SPI_p = \sum(\text{Weight}_i \times \text{Rating}_i)$$

Where $\sum \text{Weight}_i = 1.0$

7. STAKEHOLDER MANAGEMENT

Stakeholder Influence / Power (0–5)

Stakeholder Interest (0–5)

Stakeholder Impact Index (SII)

$$SII = \left(\frac{\text{(Influence + Interest)}}{2}\right)$$

Engagement Impact

Engagement Impact = SII × Engagement Level

8. COMMUNICATION MANAGEMENT

Communication Effectiveness (CE)

$$CE = \left(\frac{\text{Messages Received \& Understood}}{\text{Total Messages Sent}}\right) \times 100$$

Communication Efficiency (C_{eff})

$$C_{eff} = \frac{\text{Useful Communication}}{\text{Total Communication}}$$

Information Overload Index (IOI)

$$IOI = \frac{\text{Volume of Information}}{\text{Capacity to Absorb}}$$

9. BENEFIT MANAGEMENT

Return on Investment (ROI)

$$ROI (\%) = \left(\frac{\text{Net Benefits}}{\text{Total Investment}}\right) \times 100$$

Benefit Cost Ratio (BCR)

BCR = Total Benefits / Total Costs

Payback Period (Years)

Payback Period = Initial Investment / Annual Net Benefits

Net Present Value (NPV)

$$NPV = \sum_{t=0}^{n} \frac{(\text{Benefits}_t - \text{Costs}_t)}{(1+r)^t}$$

Where r = discount rate

10. FORMULA SUMMARY TABLE

Area	Formula	Key Use
EVM	CPI = EV / AC	Cost performance
EVM	SPI = EV / PV	Schedule performance
EVM	EAC = BAC / CPI	Forecast at completion
Schedule	TF = LS – ES	Schedule flexibility
Risk	RE = P × I	Risk exposure
Risk	EMV = ∑ (P × I)	Risk decision making
Quality	DPMO = (D / Opp) × 10^6	Process quality
Procurement	TCO = ∑ All Costs	Total ownership cost
Stakeholder	SII = (I + In) / 2	Stakeholder priority
Benefits	ROI = (B / I) × 100	Investment return

PV: Planned Value | EV: Earned Value | AC: Actual Cost | BAC: Budget at Completion
ES: Early Start | EF: Early Finish | LS: Late Start | LF: Late Finish | TF: Total Float
P: Probability | I: Impact | D: Defects | Opp: Opportunities | B: Benefits | I: Investment
∑: Summation | r: Discount Rate | t: Time Period

11. COMMON DISTRIBUTIONS (RISK)

1. Beta Distribution (PERT)

$$E = \frac{(O + 4M + P)}{6}$$

$$\text{Variance} = \left(\frac{(P - O)}{6}\right)^2$$

Where: O = Optimistic M = Most Likely P = Pessimistic

2. Triangular Distribution

$$E = \frac{(O + M + P}{3}$$

3. Standard Deviation

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{n - 1}}$$

Where μ = mean, n = number of observations

12. NET PRESENT VALUE (NPV) DETAILS

$$NPV = \sum_{t=0}^{n} \frac{(\text{Cash Flow}_t)}{(1+r)^t}$$

Cash Flow_0 = –Initial Investment

r = Discount Rate (required rate of return)

NPV > 0 → Accept

NPV = 0 → Break Even

NPV < 0 → Reject

ORACLE PRIMAVERA P6

EXAMPLE FOR MANHOURS APPORTIONING @ VARIOUS LINEAR LEVELS INSIDE EPC SCHEDULES BUILT ON P6

1 Short Description

In Primavera P6, weightage represents the relative importance or contribution of an activity, WBS or work package to the total project scope. It is used to calculate planned progress, actual progress and overall project completion. The total project weightage always equals 100 %.

2 Methods of Dividing Weightages

Method	Basis	When to Use
Cost Based	Activity budgeted cost	Most common, EPC projects
Quantity Based	Measurable quantities (m, ton, m³, etc.)	Construction / installation work
Man-Hour Based	Estimated labor hours	Labor intensive activities
Assigned / Management	Scope importance / approved rules	Milestones, lump-sum or client-defined

3 Formulas

Cost Based : $W = \frac{\text{Activity Cost}}{\text{Total Cost}} \times 100$

Quantity Based : $W = \frac{\text{Activity Quantity}}{\text{Total Quantity}} \times 100$

Man-Hour Based : $W = \frac{\text{Activity Man-hours}}{\text{Total Man-hours}} \times 100$

4 Example – Weightage Distribution (Cost Based)

WBS / Activity	Budget	Weightage
Civil Works	200,000	20 %
Mechanical Works	400,000	40 %
Electrical Works	250,000	25 %
Commissioning	150,000	15 %
Total	1,000,000	100 %

5 Weighted Progress Calculation

WBS	Weightage	Actual %	Weighted Progress (W × P)
Civil	20 %	80 %	16 %
Mechanical	40 %	50 %	20 %
Electrical	25 %	40 %	10 %
Commissioning	15 %	20 %	3 %
Total	100 %	–	49 %

Weighted Progress = $\sum(W_i \times P_i)$

W_i = Weightage of activity
P_i = Actual % complete

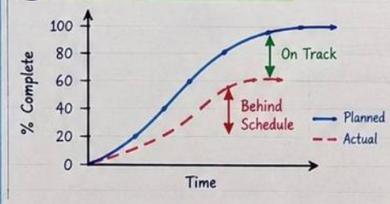
6 Steps in Primavera P6

1. **Create Project WBS** – Divide project into discipline-wise or scope-wise WBS.
2. **Develop Activities** – Add detailed activities with logic, durations and resources.
3. **Assign Budget / Resources** – Load cost or labor resources as per approved budget.
4. **Define Weightage Basis** – Select cost, quantity, man-hours or approved method.
5. **Validate Weightages** – Confirm total weightage equals 100 %.
6. **Calculate Progress** – Schedule the project, update actuals and generate planned vs. actual S-curves.

7 Key Points

- ✔ Weightage should reflect approved scope and measurement methodology.
- ✔ Avoid equal weightage for activities with significantly different costs.
- ✔ Do not double-count WBS and activity weightages.
- ✔ Use consistent rules for planned and actual progress.
- ✔ Confirm if procurement, engineering, construction and commissioning are weighted separately.
- ✔ Document the basis of weightage for progress reporting and claims.
- ✔ Weightage is not the same as activity duration or percentage complete.

8 Example S-Curve Result



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Description: PROJECT CONTROLS Important Formulae in Practice

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